Table 1: The case study uses these parameters and yields these results.

Quantity	Value
VEHICLE	
Model	CF2X (Crazyflie 2.x, X layout)
Mass, arm length	27 g, 39.7 mm
Thrust coefficient k_f	3.16×10^{-10} N/rpm^2
Torque coefficient k_m	7.94×10^{-12} N m/rpm^2
Maximum rotor speed	21 703 rpm
Thrust-to-weight ratio	2.25
SIMULATION	
Simulator	gym-pybullet-drones 2.2.0
Physics, control rate	240 Hz, 120 Hz
Integrator	Bullet, semi-implicit Euler
Drag, ground effect, downwash	off
Initial state	at rest on the ground, yaw 0
CONTROLLER	
Hover throttle, lean limit	0.444, 0.35 rad (20°)
Position gains (xy, z)	1.0, 1.5 s^{-1}
Velocity PID xy, z (k_p, k_i, k_d)	(3, 0.5, 0.25), (4, 2, 0)
Angle gains (roll, pitch, yaw)	8.0, 8.0, 4.5 s^{-1}
Rate PID roll, pitch (k_p, k_i, k_d)	0.135, 0.135, 0
Rate PID yaw (k_p, k_i, k_d)	0.2, 0.02, 0
Speed limits (xy, z)	1.0, 1.0 m/s
Acceleration limits (xy, z)	3.0, 3.0 m/s^2
MISSION	
Waypoints w_0, w_1, w_2 (NED)	$(0, 0, -1)$, $(2, 0, -1)$, $(2, 2, -1)$ m
Yaw at w_0, w_1, w_2	0°, 0°, 90°
Acceptance radius	0.05 m
Hold at w_2	3 s within 0.1 m
BUILD	
Lean toolchain	v4.33.0-rc1
C compiler	leanc (clang 22.1.4)
Compiler options	`-O3 -ffp-contract=off`
Binary size	2.54 MB
Linking	Lean runtime static
RESULTS	
Final position error	0.12 cm
Maximum |roll|, |pitch|	11.1°, 11.9°
Within 0.1 m of w_0, w_1, w_2	1.28, 4.15, 7.10 s
Tick round trip, lake (median, p99)	136, 157 µs
Tick round trip, bin (median, p99)	140, 157 µs
Identical motor commands, 3 routes	1213 of 1213 ticks